Popular Pretrained Models (Category Wise)

1. Computer Vision (Images)

Definition:
Computer Vision is a field of Artificial Intelligence that enables computers to understand, analyze, and extract information from images and videos, similar to how humans see and interpret visual data.

Examples:

- Face Recognition
- Object Detection
- Medical Image Analysis
- Self-Driving Cars

Popular Models: VGG16, ResNet50, EfficientNet, YOLO

Model	**Use Case**
ResNet50 / ResNet101	Image Classification, Feature Extraction
VGG16 / VGG19	Image Classification
InceptionV3	Image Classification
MobileNetV2	Mobile & Edge Devices
EfficientNet	High Accuracy with Fewer Parameters
EfficientNetV2	Faster Training
Vision Transformer(ViT)	Image Classification
Swin Transformer	Advanced Vision Tasks
ConvNeXt	Modern CNN Architecture
YOLOv8	Object Detection

Common datasets used for pretraining: ImageNet, COCO.

- https://www.geeksforgeeks.org/computer-vision/top-pre-trained-models-for-image-classification/
- https://docs.pytorch.org/vision/stable/models?utm_source=chatgpt.com
- https://www.analyticsvidhya.com/blog/2020/08/top-4-pre-trained-models-for-image-classification-with-python-code/

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2. NLP (Text)

Definition:
Natural Language Processing (NLP) is a branch of AI that enables computers to understand, interpret, and generate human language in text or speech form.

Examples:

- Chatbots
- Sentiment Analysis

- Language Translation
- Text Summarization

Popular Models: BERT, RoBERTa, T5, GPT

Model	Use Case
BERT	Text Classification, Sentiment Analysis
RoBERTa	Improved BERT
DistilBERT	Lightweight BERT
ALBERT	Memory Efficient BERT
XLNet	Context-Aware NLP Tasks
T5	Text Generation, Summarization
GPT Family	Text Generation, Chatbots
BART	Summarization, Translation
Sentence-BERT	Sentence Embeddings

BERT, RoBERTa, T5, and GPT are among the most widely used pretrained NLP models.

- https://en.wikipedia.org/wiki/BERT_%28language_model%29?utm_source=chatgpt.com
- https://en.wikipedia.org/wiki/T5_%28language_model%29?utm_source=chatgpt.com
- https://www.geeksforgeeks.org/nlp/top-5-pre-trained-models-in-natural-language-processing-nlp/

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3. Audio / Speech

Definition:
Audio and Speech Processing is a field of AI that focuses on analyzing, understanding, and converting human speech and audio signals into meaningful information.

Examples:

- Speech-to-Text
- Voice Assistants
- Speaker Recognition
- Voice Commands

Popular Models: Whisper, Wav2Vec 2.0, DeepSpeech

Model	Use Case
Wav2Vec 2.0	Speech Recognition
Whisper	Speech-to-Text
DeepSpeech	Speech Recognition

- OpenAI Whisper Tutorial (DataCamp)
- Hugging Face Wav2Vec2 Guide
- Mozilla DeepSpeech Documentation
- IBM Speech Recognition Overview

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4. Multimodal (Image + Text)

Definition:
Multimodal AI is an AI system that can process and understand multiple types of data such as text, images, audio, and videos simultaneously to perform tasks more effectively.

Examples:

- Image Captioning
- Visual Question Answering
- AI Assistants that understand images and text
- Document Understanding

Popular Models: CLIP, BLIP, Flamingo, Gemini

Model	Use Case
CLIP	Image-Text Understanding
BLIP	Image Captioning
Flamingo	Vision-Language Tasks

- Hugging Face BLIP Tutorial
- DeepMind Flamingo Overview
- IBM Multimodal AI Explanation

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5. Generative AI

Definition:
Generative AI is a type of Artificial Intelligence that can create new content such as text, images, audio, videos, and code based on patterns learned from existing data.

Examples:

- ChatGPT
- Image Generation

- Content Writing
- Code Generation

Popular Models: GPT, Llama, Gemini, DALL·E, Stable Diffusion

Model	Use Case
GPT Series	Text Generation
Llama Series	Open-source LLM
Gemini	Multimodal AI
Stable Diffusion	Image Generation
DALL·E	Image Generation

- IBM Generative AI Guide
- NVIDIA Generative AI Explained
- AWS: What is Generative AI?